

Industrial Predictive Quality MVP

Plain-English Sales Briefing for Audi and Industrial Manufacturers

EVOCON Solutions

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One sentence: EVOCON's Industrial Predictive Quality MVP helps factories find visual defects, machine-risk patterns and process deviations earlier by combining images, sensors and process data into a ranked inspection queue.

1 Purpose of This Document

This document is written for a non-technical salesperson or executive who needs to explain the MVP in a factory meeting. It avoids academic language and focuses on what the customer cares about:

- fewer late defects;
- fewer unexpected machine problems;
- less noise from alarms;
- clearer inspection priorities;
- a low-risk pilot that can prove or disprove value in 4–6 weeks.

2 What Problem Are We Solving?

Modern factories already collect large amounts of data: camera images, sensor traces, PLC/SCADA values, torque curves, test logs, recipes, setpoints and maintenance events. The problem is that these signals are usually reviewed separately.

The MVP turns these separate signals into one practical question:

Which part, cycle, machine, batch or line should the team review first?

For a premium automotive manufacturer such as Audi, the relevant pain points are surface quality, assembly quality, machine drift, supplier quality, line stoppages and late detection of anomalies.

3 Explain It Like I Am Five

Imagine an experienced quality engineer who sees thousands of good parts and machine cycles. Over time, the engineer develops a feeling for what normal production looks like.

The MVP does something similar, but with data:

1. it looks at images and sensor traces from normal production;
2. it learns what good production usually looks like;
3. it watches new production data;
4. it highlights what changed;
5. it ranks what should be checked first.

It does not replace engineers. It gives engineers a better review queue.

4 The Product in Plain English

Module	What it does
Visual Quality	Looks at part or product images, marks suspicious areas with heatmaps and ranks images by review priority.
Machine Risk	Uses vibration, current, temperature, torque, SCADA or test logs to score whether a machine/cycle is becoming risky.
Process Deviation	Compares how a process should evolve with how it actually evolves, then marks the first point where it starts drifting.
Hierarchical Ranking	Aggregates local evidence from patch to image, part, cycle, batch and line, so managers see useful priorities instead of raw signals.

5 Why This Fits Audi

Audi and Volkswagen Group plants have complex production systems with high quality expectations. The MVP is relevant wherever the plant has:

- image-based inspection of parts, surfaces, welds, paint, connectors or assemblies;
- torque, vibration, current, temperature or test-log traces;
- station-level or line-level quality events;
- process settings such as recipe, speed, force, pressure, temperature, material, tool or robot program;
- a need to reduce manual review burden or identify the first warning signs of drift.

6 Suggested First Audi Use Cases

Use case	Data needed	Output
Body / paint / surface inspection	Images from good and questionable parts, station ID, inspection results	Heatmap and part review ranking
Assembly station risk	Torque curves, press-fit traces, robot/PLC events, station ID	Top risky cycles and deviation timeline
Tool or machine drift	Vibration/current/temperature, maintenance events, process context	Machine risk score and early-warning ranking
Supplier quality	Incoming-part images, batch/supplier metadata, inspection labels	Batch and supplier review prioritisation
End-of-line test anomaly	Test logs, final pass/fail labels, process context	Test anomaly score and root-candidate ranking

7 What Results Can We Honestly Show?

Current internal MVP results support selling a pilot, not a production guarantee.

Area	Current evidence	Commercial meaning
Visual anomaly detection	DINOv2/ResNet + PatchCore/PaDiM achieved near-perfect quick-run scores on MVTec bottle.	Visual inspection is the strongest first product module to pilot.
Sensor failure-soon risk	Engineered sensor features are strong; future learned signal is promising but not a replacement.	Start with reliable physical indicators, then add learned signals if they improve validation.
No-cycle sensor evidence	Sensor and latent signals help when lifecycle/cycle proxies are removed.	Useful when simple counters do not explain risk.
Hardness/material generalization	Sensor features were strong in held-out hardness/material-style splits.	Physical signals can matter when process regime changes.

8 What Not To Claim

Do not say:

- “This is production-ready tomorrow.”
- “This replaces quality engineers.”
- “The model guarantees zero defects.”
- “The research model always beats engineered features.”
- “We have already deployed this at Audi.”

Say instead:

“We have a pilot-ready MVP. We can take one process, compare against strong baselines and show in 4–6 weeks whether images, sensors and process data can detect defects or drift earlier than current checks.”

9 The Recommended Meeting Script

Opening

“EVOCON already works in industrial automation and project support. This MVP adds an AI layer for predictive quality: it combines camera inspection, sensor trends and process context to produce a ranked review queue. We are not asking you to replace your systems. We propose a 4–6 week pilot on one line to measure whether we can catch defects or machine drift earlier.”

First Question

“Where do you currently detect quality or machine issues too late?”

If They Mention Visual Quality

“Then the first pilot should be visual: part images in, heatmaps and inspection ranking out. We compare against simple rules and strong visual baselines.”

If They Mention Machine Downtime

“Then the first pilot should use sensor/process data: vibration, current, temperature, torque, SCADA or test logs. We rank cycles or machines by risk and measure lead time and false alarms.”

If They Mention Too Many Alarms

“Then the first pilot should focus on prioritisation: the top 5–10% alerts that deserve human review, with evidence for why they were selected.”

10 What Data To Ask For

Minimum	100–1,000 historical images or sensor windows from one process, with timestamps and any known quality/failure labels.
Better	Images + sensor traces + process settings + station IDs + inspection results + maintenance/quality events.
Best	A full sequence: part/cycle ID, timestamp, station, line, recipe/setpoint, image, sensor trace, inspection result and downstream event.

11 4–6 Week Pilot Plan

Week	Work
1	Data audit, target definition and baseline agreement.
2	Build simple baselines and visual/sensor foundation models.
3–4	Train/evaluate risk ranking, heatmaps, deviation timelines and calibration.
5	Review with plant experts; collect false positives, missed cases and practical constraints.
6	Deliver executive report, local demo, metrics, go/no-go recommendation and integration roadmap.

12 What EVOCON Should Send Before the Meeting

- The ELI5 HTML/PDF deck: [audi_industrial_predictive_quality_eli5_en.html/pdf](#).
- The algorithm animation: [assets/industrial_predictive_quality_algorithm_demo.html](#).
- A short email, not the full technical paper.
- Offer a 15-minute qualification call before asking for data.

13 Short Email Template

Subject: 4–6 week pilot for earlier quality and machine-risk detection

Hello [Name],

EVOCON Solutions works in industrial automation and project support. We are preparing a pilot-ready Industrial Predictive Quality MVP for automotive manufacturing: it combines camera inspection, sensor trends and process context to produce heatmaps, risk scores and a ranked review queue.

The proposal is deliberately narrow: choose one line/process, use historical images or sensor/test data, compare against strong baselines and deliver a go/no-go report in 4–6 weeks.

Would it make sense to have a 15-minute call to identify one quality or machine-risk case where late detection is costly?

Best regards, [Sender]

14 Final Positioning

The strongest commercial message is:

We turn existing factory data into earlier warnings and clearer actions.

The safest commercial ask is:

Let us validate one process in a 4–6 week pilot.